Girls Build Bots — Brownie Automotive Design

Badge-aligned workshop guide for robotics-team-led outreach

Level: Brownies (Grades 2–3) · **Badge:** Automotive Engineering — Automotive Design **Series:** Automotive Engineering (Design · Engineering · Manufacturing) **Format:** One 75–90 minute session · **Ratio:** 1 mentor per 3–4 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on craft and cardboard supplies the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie Automotive Engineering badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: designing for a driver

The Automotive series follows a car from idea to road: **Design** it, **engineer** it to move, then **manufacture** it. Across all three, Brownies learn that engineers build for a *person* — a driver with real needs. Mentors: keep asking "who is this car for?"

Vocabulary: design, vehicle, wheel, chassis (the frame), passenger, sketch, model.

Automotive Design (75–90 minutes)

Badge theme: Thinking like a car designer — who the vehicle is for, what it needs to do, and how it should look — then sketching and modeling one.

Materials

- Cardboard, construction paper, bottle caps (wheels), craft sticks, tape, markers
- "Customer cards" (a farmer, a dog-walker, a pizza deliverer, a family with three kids)
- Design sheet: My driver / What they need / My car's special feature

Activity 1 — Cars Have a Job and a Driver (15 min)

Look at different vehicles (fire truck, ice-cream truck, race car, minivan). For each: **who drives it, and what does it need to do that job?** A fire truck needs a big ladder; an ice-cream truck needs a freezer. Cars are designed *around* their driver's needs.

Activity 2 — Meet Your Customer (15 min)

Each Scout (or pair) draws a **customer card**. They fill the design sheet for that specific driver: who they are, what they need, and one special feature their car will have. Designing for a real customer — not just "a cool car" — is the heart of the badge.

Activity 3 — Sketch, Then Build a Concept Model (40 min)

Designers sketch before they build. Scouts draw their car, then build a cardboard **concept model** with wheels that turn. It doesn't have to drive — it has to show the design and its special feature for that customer.

Activity 4 — Design Show (10 min) + Wrap (5 min)

Each Scout shows her car and names the driver and the special feature in one sentence ("This is for a dog-walker, so it has a comfy pet seat!"). Gallery walk with one kind comment each.

> **Robot moment (optional):** Share for 60 seconds how the team designs its robot around the year's challenge — deciding what it needs to *do* before deciding what it looks like, just like designing a car for a driver.

Completion Checklist — Automotive Design

By the end of the session, to earn the badge, each Scout can: - [ ] Explain that vehicles are designed around a driver's needs - [ ] Describe a specific customer and what their car needs to do - [ ] Sketch a car design before building it - [ ] Build a concept model with a special feature and explain who it's for

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Mentor Guide (for the robotics team)

- **You don't need a robot for this.** It runs on cardboard and craft supplies. The "robot moment" box is a 60-second add-on if you brought one.
- **Push the "who."** Keep steering Scouts back to their customer — the design should fit that driver.
- **Sketch first, always.** Gently insist on a drawing before the build; it's a real design habit.
- **Function over neatness.** Praise the idea and the fit-to-customer, not how tidy the model is.
- **Watch the clock.** Brownies fade after ~25 minutes on one thing; keep segments moving.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.

3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Brownie Automotive Engineering badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.